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Stock Market Prediction using Social Media Analysis

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Abstract

Stock Forecasting is commonly used in different forms everyday in order to predict stock prices. Sentiment Analysis (SA), Machine Learning (ML) and Data Mining (DM) are techniques that have recently become popular in analyzing public emotion in order to predict future stock prices.

The algorithms need data in big sets to detect patterns, and the data has been collected through a live stream for the tweet data, together with web scraping for the stock data. This study examined how three organization's stocks correlate with the public opinion of them on the social networking platform, Twitter.

Implementing various machine learning and classification models such as the *Artificial Neural Network* we successfully implemented a company-specific model capable of predicting stock price movement with 80% accuracy.

Keywords: Statistical Learning; Artificial Intelligence; Neural Network; Machine Learning; Support Vector Machine; Twitter; Stock Forecasting.

Referat

Aktieprisprognos är dagligen använt i olika former för att kunna förutspå aktiekurser. Opinionsanalys (O), Maskininlärning (ML) och Data Mining (DM) är tekniker som har blivit populära för att kunna mäta och analysera folkopinionen och därmed förutsäga framtida aktiekurser.

Algoritmerna behöver data i stora mängder för att kunna känna igen mönster. Data från Twitter har insamlats genom en realtidsström, medan aktiedatat har samlats in via webbskrapning. Detta arbete har examinerat hur tre organisationers aktie korrelerar med folkopinionen på den sociala nätverksplattformen Twitter.

Efter att ha implementerat maskininlärnings- och klassifikations modeller såsom *Artificiella neuronnät* har vi implementerat en modell som är kapabel till att förutspå ett företags akties prisrörelse med 80% noggrannhet.

Nyckelord: Statistiskt lärande; Artificiell intelligens; Neurala nätverk; Maskininlärning; Stödvektormaskin; Regressionsanalys; Twitter; Aktieprisprognos.

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Chapter 1

Introduction

Stock forecasting, or stock market prediction is a common economic activity that has been an attractive topic and issue to researchers of engineering, finance, computer science, mathematics and several other fields. The prediction of stock markets is considered to be a challenging task of financial time series prediction. Stock forecasting is challenging by nature due to the complexity of the stock market with its noisy and volatile environment, considering the strong connection to numerous stochastic factors such as political events, newspapers as well as quarterly and annual reports (Ticknor 2013).

News and updates are unpredictable and it is of great interest to evaluate if there is a relationship between an organization's stock and the public emotion. One approach is to analyze the public emotion of an organization in order to forecast the progress of the organizations' stock.

Analysis of social media activity is strongly related to *Sentiment Analysis* which is commonly used in many industries and provides stakeholders with great tools for understanding how the common person reacts to certain events (Thovex and Trichet 2013; Castellanos et al. 2011). *Opinion Mining* and *Sentiment Analysis* is conducted via different methodological approaches. Many of these approaches are spawned from *Natural Language Processing* or make use of data mining methodologies such as *N-grams* (Arafat, Ahsan Habib, and Hossain 2013).

The use of *Supervised Sentiment Analysis* have in previous research been used as a predictor of stock price movement with promising results (Makrehchi, Shah, and Liao 2013). Furthermore optimized *Artificial Neural Network* algorithms have been proved to successfully predict the stock market with a reasonably low percentage of error on financial data. Such data includes stock opening price, close price and trade volume (Ticknor 2013).

Existing research on the subject strongly focuses on stock forecasting solely, taking economical key values gathered from various financial resources and stock trends into consideration.

1.1 Problem statement and hypothesis

The purpose of this thesis is to analyze if social media analysis can be used to predict a company's stock price. The following problem to be investigated is therefore *can social media analysis be used solely to predict a company's stock price?*

We expect that social media has a strong impact on a company's stock price. However, we are unsure if the impact is strong enough to be used exclusively for stock market prediction.

1.2 Scope and objectives

In this paper we have investigated the possibilities of analysing social media with *Machine Learning* and *Sentiment Analysis* for stock market forecasting.

Previous research in the field of social media analysis and *Sentiment Analysis* have mostly focused on the gathering of public data through blogs rather than on social media and Twitter. In recent research Facebook and Myspace data have been extracted and analyzed in the same manner (Arafat, Ahsan Habib, and Hossain 2013). The scope of this thesis is limited to analysis of Twitter data from three different companies in three different industries.

Chapter 2

Background

This background will consist of four different parts. The first part covers statistical learning and corresponding methods, followed by the fundamentals of stock forecasting using AI and thereafter an introduction and discussion on *Opinion Mining* and *Sentiment Analysis*. At last a section discussing the advantages and disadvantages on analysing Twitter as a platform and the potential of using the result as a valuable asset and resource for financial investments.

2.1 Statistical Learning

In this chapter the statistical learning methods applied on the data will be presented. This chapter will include definitions, explanations and examples of the models and concepts that have been used throughout this thesis. *Artificial Intelligence*, *Machine Learning* and *Sentiment Analysis* are introduced along with elementary statistical concepts. The elementary concepts are followed by deeper explanations of relevant approaches.

2.1.1 Regression Analysis

In mathematical statistics, regression is the prediction of qualitative data relationships. The aim of regression analysis is to model the relationships between one or more dependent or independent variables. These relationships are not necessarily of equal strength but more commonly of varying strength.

The most simple form of regression analysis is *linear regression* constrained by the assumption that there is a linear relationship between given variables. There are various methods to estimate the variable values, where the most basic method is the simple linear regression

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where β_0 and β_1 represents the parameters, x_i the independent variable and ϵ the error term.

The linear model is easily developed by adding additional variables to the equation, such as assuming a parabola function.

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_1^2 + \epsilon_i$$

Regression analysis most commonly use the mean squared error to predict how well the linear regression model performed. The residuals of the model is the difference between the true value y_i and the predicted model value $\hat{y}_i$.

$$\epsilon_i = y_i - \hat{y}_i$$

The sum of squared residuals SSE is calculated as

$$SSE = \sum_{i=1}^{n} y_i - \hat{y}_i$$

The Mean Squared Error MSE is calculated dividing the SSE with the number of observations

$$MSE = \frac{1}{n} \sum_{i=1}^{n} y_i - \hat{y}_i$$

alternatively

$$MSE = \frac{SSE}{n}$$

Using a multiple regression analysis on the three stock variables open, close and high price of the month, researchers established a model with a 89% accuracy on predicting stock price movement (Kamley, Jaloree, and Thakur 2013).

2.1.2 Artificial Intelligence

Artificial intelligence (AI) is a scientific field which strives to *build* and *understand* intelligent entities. Existing formal definitions of AI address different dimensions such as *behaviour*, *thought processing* and *reasoning*. The distinguishing between *human* and *rational* behaviour is often mentioned in the field. To create AI the two components intelligence and tools are required. Computer Science have created such tools (Russel and Norvig 2009).

2.1.3 Data Mining

Data mining is the process of extracting information from large data sets (commonly known as big data) in order to predict trends, behaviour and other types of information that serve as a foundation for the organizations capability to make data-driven decisions. Extraction of previously unknown and potentially useful information from existing databases is an effective way of data mining, commonly referred as knowledge discovery, or KDD (Das and Shorif Uddin 2011).

Little research has been done on the efficiency of data mining solely as a stock predictor (Das and Shorif Uddin 2011) but the use of integrated data mining techniques such as dynamic time series, ANN and Bayesian probability has been proven both reliable and useful (C. Huang and Lin 2014; Das and Shorif Uddin 2011).

2.1.4 Machine Learning

Machine learning (ML) is a subfield of AI concerned with the implementation of programs and algorithms that can learn autonomously (Russel and Norvig 2009). Machine learning has strong connections with statistical and mathematical optimization, whereas all of these areas aim at locating interesting regularities, patterns and concepts from empirical data. Therefore, statistics and mathematical optimization provide methods and applications to the area of machine learning (Hand, Manilla, and Smyth 2001).

A major issue and drawback for the use of ML and classification models is the risk of overfitting, which is when a learning algorithm overestimates the parameters in the training data. overfitting could lead to low precision on unknown data, as it tries to generalize what was learnt from the training data. This very reason is why hybrid techniques are of great interest for researchers, since this decreases the risk of overfitting and increases the chance of more accurate weights and models (Ticknor 2013).

2.1.4.1 Representation of Data

The performance of AI and ML methods heavily rely on the representation of the data. The design of preprocessing pipelines and data transformation are important for the deployment of the ML methods. Therefore, the data representation is dependent on being expressive; it should encapsulate a big variation of the data without significant information being left out. In order to get the best result possible of an ML application, the data needs to be selected carefully (Bengio, Courville, and Vincent 2014). In this thesis, the data has been carefully chosen with concern to different industries and gathered through a real-time feed to get the most accurate, unbiased and expressive variation possible of the data.

2.1.5 Natural Language Processing

Natural language processing (NLP) is a field in artificial intelligence and linguistics concerned with interaction between computers and human natural language. As a part of *Human-Computer Interaction* NLP is concerned with enabling computers to derive and interpret human natural language. Recent work in NLP are algorithms based on ML and more specifically statistical machine learning (Russel and Norvig 2009).

State of the art applications of NLP consist of text classification, information extraction, sentiment analysis, machine translation and is applied to many different scientific areas (Google 2015). Some of these areas are biomedicine (Doan et al.

2014) and economics. Discussed in depth in chapter 2.2-2.4, *Sentiment Analysis* approaches have been applied to stock forecasting and financial modelling (Arafat, Ahsan Habib, and Hossain 2013).

2.1.6 Correlation and Causality

Statistical learning is derived from mathematical statistics. In mathematical statistics, the term *correlation* is commonly used. *Correlation* is defined as how strong a connection is between two events and the direction of the connection.

The covariance of two events is defined as follows,

$$\sigma(X, Y) = \mathrm{E}\left[(X - \mathrm{E}[X])(Y - \mathrm{E}[Y])\right],$$

where E[X] is defined as the *expected value*. If for two events X and Y, $\sigma(X, Y) = 0$, then X and Y are uncorrelated.

The *correlation*, or *dependence* is defined mathematically as,

$$\rho_{X,Y} = \frac{\sigma(X,Y)}{\sigma_X \sigma_Y}, \sigma(X, Y) \in [-1, 1],$$

where σ represent the standard deviation (Blom 2004). $\rho_{X,Y}$ is in the domain [-1,1] and the outcomes are as follows,

$\rho_{X,Y} = 1$
: Represents the maximal positive connection between X and Y, and they are in the exact same directions. If X is going upward, Y will go upward too.

$\rho_{X,Y} = -1$
: Represents the minimal negative connection between X and Y, and they are in completely different directions. If X is going upward, Y will go downward.

The correlation between two events do not necessarily imply that one of the events have caused the other.

Causality is defined as that the events have caused each other. In statistics, pre-existing data or experimental data is employed to infer causality by regression methods. When analyzing a casual inference, the main task is to distinguish between association and causation. *Association* describes situations where situations occur more often together, and vice versa. The associations do not need to be meaningful, and is due to the expectation that they reflect a casual relation.

Associations can be observed without an underlying casual relation, and a cause X together with a response Y will be associated if X is indeed casual for Y, but necessarily not vice versa.

The following conclusions and cases exist for two correlated events X and Y:

1. X causes Y
2. Y causes X
3. X and Y are results from another event, but do not cause each other
4. There exist no correlation between X and Y, they are random events

(Blom 2004).

In this thesis, the correlation between the stock market and the corresponding social media posts for three specific organizations and industries will be analyzed. These four events will be considered when analyzing our results.

2.1.7 Supervised Machine Learning

Supervised machine learning aims to predict output data sets $(y_1, y_2, ..., y_n)$ from given sets of input data $(x_1, x_2, ..., x_n)$ for n observations. A general machine learning function is created for predicting output from the input that has not been a part of a training set. The predictions are formed by a training set of tuples $((y_1, x_1), (y_2, x_2), .., (y_n, x_n))$ from a known set of input and output.

There are different types of output and they can be divided into two types of prediction problems, *classification* and *regression*. Classification and regression have a lot in common, but there are specific learning algorithms that are specialised for each method.

2.1.7.1 Classification

In terms of ML, classification is the process of identifying which category a given data object belongs to. Supervised machine learning requires a set of correctly classified data and use it to train a classifier for classification of non-classified data instances.

Each data entry is analyzed separately and evaluated on an attribute basis in order to predict the correct classification category.

2.1.7.2 Decision Tree

A *Decision Tree* is a data mining technique designed to create a model predicting the outcome value of a target variable based on the value of all the other input variables.

The tree is built using nodes, edges and leafs. Every node in the decision tree corresponds to an input variable where the edges between these represent possible

values of the input variables. The leaf nodes corresponds to the deducted target variable value based on the path from root to leaf.

The most commonly used learning algorithm is the *top-down induction* recursive partitioning algorithm. This algorithm is greedy as it makes optimal choices at each recursive step.

The ID3 decision tree learning algorithm is commonly used and is based on the concept of *information gain* and *entropy* which measures the unpredictability or impurity of the content. Information gain is informally described as

$$Information\ Gain = entropy(parent) - [average\ entropy(children)]$$

The *ID3* algorithm splits on the attribute with the highest information gain to reduce the impurity of the content as much as possible.

2.1.7.3 Random Tree

The *random tree* is a functions in the exact same way as the *decision tree* but the fact that only a subset of attributes are available after each split. This approach provides more resilience to noise (Li et al. 2010).

2.1.7.4 Support Vector Machine

Support Vector Machines (SVM) or *Support Vector Networks* (SVN) are classification and regression analysis techniques. Support vector machines are supervised learning models for data analysis and pattern recognition. Common application areas are image recognition, text analysis and bioinformatics. The support vector machine constructs a hyperplane, or a set of hyperplanes in a high- or infinite-dimensional space.

In many cases, the data is not linearly separable. Using a SVM learning algorithm, it is possible to create a transformable room. The model represents the examples as points in space, maps separate categories and divides them as much as possible. The goal is to design a hyperplane that classifies all training vectors into two distinct classes, where the best choice is the hyperplane that leaves the maximum margin from both classes. (Platt 1999; Microsoft 2015)

Recent research and state of the art approaches of Support Vector Machines shows that using ensemble approaches can drastically reduce the training complexity, while maintaining high predictive accuracy. This has been done by implementing the SVMs without duplicate storage and evaluation of support vectors, which has been shared between consistent models. The approach used with the software EnsambleSVM uses a divide-and-conquer strategy by aggregating multiple SVM models, trained on small subsamples of training sets. For p classifiers on n/p subsamples, the approximate complexity of $\Omega(n^2/p)$. (Claesen et al. 2014)

2.1.7.5 Naïve Bayes Classifier

The *Naive Bayes* methods are a set of supervised learning algorithms that is used for clustering and classification (Lowd and Domingos 2005). The methods are based on applying Thomas Bayes' theorem with a *naive* assumption of independence between every pair of features. The naive bayes classifiers are linear classifiers and are simple, perform well and are very efficient (H. Zhang 2004; Raschka 2014). For small sample sizes, naive Bayes classifiers can outperform more powerful alternatives. However, non-linear classification problems can lead to poor performances of naive Bayes classifiers. These methods are used in a various of different fields such as diagnosis of diseases, classification of RNA sequences in taxonomic studies and spam filtering in e-mail clients (Raschka 2014).

Research of Naive Bayes have previously been proved the methods to be an optimal method of clustering and classification, no matter how strong the dependencies among the attributes are. If the dependencies distribute evenly in classes or if they cancel each other out, Naive Bayes performs optimally. (H. Zhang 2004).

Recently, Naive Bayes theorem have been applied to image classification algorithms, where the Local Naive Bayes Nearest Neighbor algorithm increases classification accuracy and improves its ability to scale to bigger numbers of object classes. The local NBNN has been shown that it is up to a 100 times speed-up over the original NBNN on the Caltech 256 dataset. (Lowe 2012)

2.1.8 Unsupervised Machine Learning

Unsupervised machine learning is the process of classifying data without access to labelled training data. Using n observations of data $(x_1, x_2..., x_n)$ the primary goal of the unsupervised machine learning method is to gather data with similar attributes and relationships into different groups. As labelled data is not provided, unsupervised methods usually require larger amounts of training data to perform equally as good as supervised machine learning methods.

2.1.8.1 K-means clustering

K-means clustering is a centroid-based clustering algorithm where k numbers of clusters is specified prior to partition of the n observations. The aim is to attach each observation to the nearest centroid.

Given n number of observations with d number of attributes forming a $d-$ *dimensional* vector, the k-means algorithm uses *Euclidean Distance* to gather similar data to each other. The objective is to minimize the within-cluster sum of squares ($WCSS$). The $WCSS$ is calculated mathematically as:

$$(d_i, \mu_i) = \sum_{j=1}^{d} (x_{i,j} - \mu_i)^2$$

where mu_1 is the mean of points assigned to cluster i.

The algorithm for k-means clustering follows the following pattern.

```
while Centroid positions are not fixed do
    Assignment: For each data point, assign it to the nearest centroid in
    terms of WCSS.
    Update: Recalculate the position of each centroid with the connected
    data points in consideration.
end
```

2.2 Stock Forecasting using Machine Learning

Stock Forecasting is one of the most common areas where Artificial Intelligence is applied, and counted for 25.4% of the total use in 1988-1995 (Wong, Bodnovich, and Selvi 1997). Earlier approaches on Stock Prediction used non adaptive programs, which have been proven to be useful for private investors placing medium-term investments. Non adaptive programs offer limited reliability for large-scale investors, since they make the most profit from short-term, large-scale transactions with low profit margin (Schoeneburg 1990).

2.2.1 Artificial Neural Networks

Most papers on Stock Forecasting take use of various Artificial Neural Networks or combinations of Artificial Neural Networks with other types of techniques, such as Bayesian regularized ANN. This is due to the nonlinear nature of the stock market, where Neural Networks are preferred. These are preferred due to their ability to deal with nonlinear relationships, fuzzy and insufficient data, and the ability to learn from and adapt to changes in a short period of time (Das and Shorif Uddin 2011). Kunwar and Ashutosh proved that the use of Neural Networks for Stock Market forecasting outperformed Statistical Forecasting methods using the 'Learn by Example' concept and furthermore proved that Neural Networks served as very good predictors for stock market prices (Kunwar and Ashutosh 2010).

ANN are commonly constructed in layers where each layer plays a specific role in the network and contains a number of artificial neurons. Typically these layers are the input layer, the output layer and numerous hidden layers in between as described in figure 1. The actual computation, processing and weighting of the neurons is done through the hidden layers and is crucial for the performance of the network (Olatunji et al. 2011).

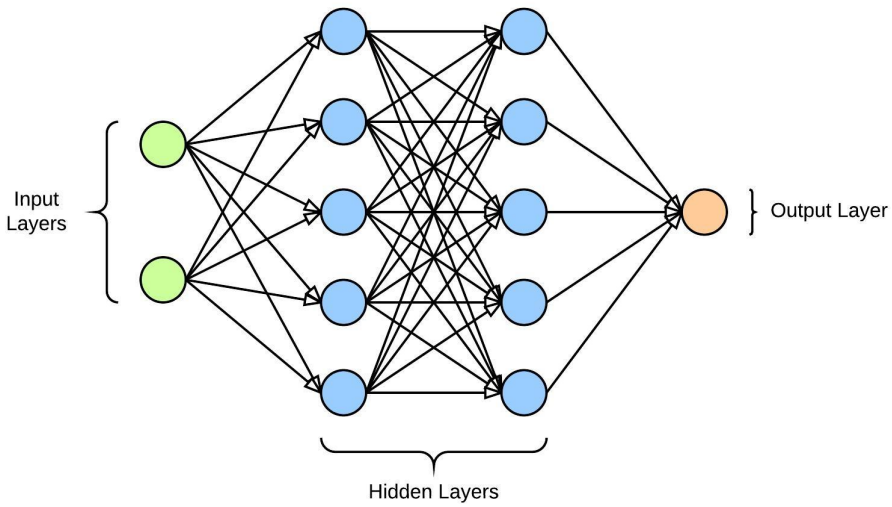

Figure 2.1. Architecture of a feedforward neural network.

2.3 Opinion Mining and Sentiment Analysis in Social Media

Sentiment Analysis refers to the automatic detection of emotional or opinionated statements in a text statement. Previous research on the subject primarily focus on reviews, which is considerably easier in terms of opinion mining, in comparison to the informal communication on social platforms (Paltoglou and Thelwall 2012).

The complexity of Sentiment Analysis and Opinion Mining on data from Social Media is due to the non-standard linguistics, heavy use of emoticons (misused punctuation), emoijis (Unicode standard characters, font), slang and incorrect grammar. Research has found that 97% of comments on MySpace contain non-standard formal written English (Thelwall 2009). Furthermore, supervised Machine Learning approaches used on reviews are problematic with Social Media data due to the lack of training data.

Classification of review data is simple because of the rating system often used, which directly classifies the review as "good" or "bad", which serves as a great source of pre-classified training data. Classified training data for Social Media would require extensive human labour for classification by hand and is thereby hard to come by, especially the quantity which would be required for good accuracy (Paltoglou and Thelwall 2012). Due to these constraints unsupervised Machine Learning algorithms have been applied using lexicon-based approaches, i.e. corpuses. These have been proven to be both reliable and robust (Paltoglou and Thelwall 2012), and

therefore an equivalent choice to the supervised approaches.

One of the main concerns on aggregating findings from Sentiment Analysis in Social Media is the assumption that these findings would be representative for the entire population of concern. Even though this might not necessarily be the case, analysis on the subject has shown a clear and consistent correlation between the results from the Sentiment Analysis and the more traditional mass survey (Ceron et al. 2009).

2.3.1 Accuracy of Sentiment Analysis

In its current stage, automated SA is not able to be as accurate as human analysis. The automated sentiment analysis methods do not account for subtleties of sarcasm, human body language or tone. In human analysis, the inter-rater reliability plays a significant part, which is the degree of agreement among raters. According to recent studies, the human agreement rate in sentiment analysis are around 79-80%. (Pak and Paroubek 2010; Wiebe, Wilson, and Cardie 2005; Ogneva 2010).

2.4 Twitter Analysis

In order for any platform to be viable as a Stock Predictor the platform itself must be suitable for data gathering. Twitter offers a comprehensive search API, up to seven days back in time, but also offers the opportunity to query against tweets in real-time, through its streaming API (Arafat, Ahsan Habib, and Hossain 2013). The Twitter API is convenient since it removes the need to batch data gathering and management, and offers an whole new aspect to Stock Prediction due to the high accessibility of data.

A major drawback using the Twitter Search API is the limitation on complexity where overly complex queries are restricted, and the limitation on availability of data older than a set number of days, seven days to be precise. This is due to the fact that the Search API makes use of indices that only contains the most recent or popular tweets, according to the Developers Page on the Twitter Website (Twitter 2015). Furthermore, it is explained that the Twitter Search API should be used for relevance and not completeness and that some tweets and users might be missing in the query results.

The Twitter Search API Developers Page propose that the Streaming API is more suitable for completeness-oriented queries which would be the case of gathering data for the Sentiment Analysis where high completeness is required to analyze the whole picture rather than specific chunks of data (Twitter 2015). The Streaming API is also favoured by existing research on the subject (Choi and Varian 2012).

Chapter 3

Methods

This chapter describes used research methods and data collection approaches. Furthermore, the methods used for Sentiment and Data Analysis are described.

3.1 Literature Study

Through research in academic articles, digital articles, papers and books within the area of machine learning and computer science, the theoretical principles of the field have been analyzed. The literature used has been accessed via the KTH library database using relevant keywords such as machine learning, sentiment analysis, artificial neural networks, stock market, stock market prediction, stock market forecasting and statistical learning. Furthermore the official Twitter documentation pages have been used. Finally, corporate information regarding our three companies of choice has been fetched through their official websites.

3.2 Data collection

To be able to implement the sentiment analysis methods and take use of statistical learning methods, adequate data sets for tweets and stocks is necessary. A significant part of this work has therefore been the collection of Twitter and financial data.

3.2.1 Twitter data collection

The Twitter data has been collected through the use of the streaming API provided by Twitter Inc. and stored in a MongoDB database. To ensure a broad and diverse set of companies to be analyzed the companies *Microsoft*, *Netflix* and *Walmart* were tracked. A brief introduction of these companies can be found in the following section.

All of the keywords used for data gathering are attached in the appendix section.

3.2.1.1 Microsoft

Microsoft is an American multinational corporation that develops, manufactures, licenses, supports and sells computer software, personal computers, consumer electronics and services. Microsoft's primary field of interest is computer software (Microsoft 2015).

3.2.1.2 Netflix

Netflix Inc. is a provider of on-demand Internet streaming media in various countries. Netflix is available in over 50 countries, and is constantly expanding. Netflix expect to be available worldwide over the next two years (Forbes, 2015). Netflix primary industry is therefore Internet services.

3.2.1.3 Walmart

Walmart is a retail company focusing on selling nutrition, but also various other kinds of products, such as medicine, clothing and electronics (Walmart 2015). Walmarts primary interest is the retail industry.

3.2.2 Java

The programming language that has been used for collecting the data through the Twitter Streaming API is Java. Java is an object oriented, platform independent and flexible general purpose language. The choice to use Java was primarily due to its flexibility and availability on different platforms. Additionally the ease to export executable Java Archives (JAR) including external libraries runnable via the terminal outperformed other alternatives.

3.2.2.1 Twitter4j

Twitter4j provides simplicity and ease when connecting to the Twitter API and gathering data. Twitter4j provides predefined functions for establishing the HTTP connection, as well as the ready-to-use implementation of listeners. Therefore, the collection of data from Twitter has been simple. Due to Twitters restricted amounts of calls to its API, three different API keys for collecting the data has been used to evade the API timeouts.

3.3 Stock data collection

The stock data has been collected using web scraping, which is the act of extracting information from the web. The web scraping method used is manual copy and paste, as the data has been collected manually from Yahoo! Finance.

Presented in tabular form below is sample stock data prices for each company, web scraped from Yahoo! Finance.

Table 3.1. Web scraped data from Yahoo! Finance

date	open	high	low	close	volume	company
4/10/2015	41.63	41.95	41.41	41.72	27,852,100	microsoft
4/9/2015	41.25	41.62	41.25	41.48	25,664,100	microsoft
4/8/2015	41.46	41.69	41.04	41.42	24,603,400	microsoft
4/10/2015	80.86	81	80.55	80.65	5,480,300	walmart
4/9/2015	80.84	81.39	80.58	80.84	3,914,600	walmart
4/8/2015	80.39	81.23	80.36	81.03	6,681,800	walmart
3/26/2015	417.4	423.13	415.73	418.26	2,285,900	netflix
3/25/2015	438.79	438.84	421.71	421.75	3,084,800	netflix
3/24/2015	427.95	441.69	427.83	438.28	2,409,500	netflix

3.3.1 MongoDB

MongoDB is a NoSQL, non-relational database for storing large amounts of data. A MongoDB database holds a set of collections, whereas a collection holds a set of documents. A document is a set of key-value pairs, much like a hashmap or a dictionary. The document data model MongoDB uses is JSON. JSON allows the user to store data of any structure and dynamically modify the schema.

3.3.1.1 Database Schema

The JSON data schema for a document is presented below.

```
{
    "_id": {
        "$oid": "55117b3577c879dc2d84a14d"
    },
    "user_name": "BasedYoona",
    "tweet_followers_count": 1558,
    "user_location": "Los Angeles",
    "created_at": {
        "$date": "2015-03-24T14:56:53.000Z"
    },
    "language": "en",
    "tweet_mentioned_count": 0,
    "tweet_ID": 580382783169695744,
    "tweet_text": "cant fucking log in 2 skype so i reset
    my password and it only resets it for microsoft account
    and not my skype what the fuck help!@?!?",
    "company": "Microsoft"
}
```

Listing 3.1. Database document sample

In this example tweet, a user is facing difficulties with his Skype account, which is a service delivered by Microsoft. As seen, a lot of swearing, slang and abbreviations are used in the tweet. With sentiment analysis with a scale ranging from 5 to -5, where every positive word has a point of 1 and every negative word a point of -1, this tweet would have been classified as -2, for the use of the negative words *fuck* and *fucking*.

Every document inside of the collection holds nine attributes, as presented in table 3.2.

3.3.2 R

R is a programming language commonly used for statistical computing and computer graphics. R is extensively used by data miners and statisticians for data analysis. The reason why R was chosen for computing the data was primarily its powerful tools and large community. R is easy to use and provides all the required functionality to perform the data analysis features necessary for stock market forecasting and SA. R is open source, and provides a big number of packages.

Table 3.2. Database attributes and their descriptions

Attribute	Description
_id	Automatically generated unique id for the document inside of the collection.
user_name	The user name for the user who posted the tweet.
tweet_followers_count	The number of followers for the user who posted the tweet.
user_location	The manually entered user location for the person who posted the tweet.
created_at	The timestamp for when the tweet was created.
tweet_ID	The unique ID for the tweet itself.
tweet_text	The actual tweet in HTML-formatted text.
company	The company that the tweet belongs to, corresponding to the search values (keywords) for the tweet.

3.4 Data preprocessing

The need for extensive data preprocessing when conducting stock market forecasting is mentioned in earlier research on the subject (Piramuthu 2006; Kaastra and Boyd 1996). Cleansing, preparation and aggregation of the collected Twitter and stock financial data was therefore required. The following section describes the preprocessing steps on the used data.

3.4.1 Data cleansing

As Twitter suffers from daily and long term spam accounts cleansing of captured data was required to ensure data quality (Thomas et al. 2011).

As retweets contain the same content as the original tweet and therefore not spam, only multiple tweets with the same content by the same author were classified as spam. This minor set of spam classified tweets were removed from the data set accordingly.

3.4.2 Sentiment Analysis

The Twitter data was collected to a MongoDB database and exported to a csv file format for further work in R. This was done through the MongoDB shell with the command *mongoexport*.

The command in listing 3.2 is used to export MongoDB data to a *csv* file. The command was executed three times in order to export each of the collections to a *csv* file format, together with the correct parameters for each of the collections.

```
mongoexport --host localhost --db dbname --collection
name --csv --out text.csv
```

Listing 3.2. Mongoexport command syntax

The SA dictionary used is presented in Appendix B and represents every negative word with a sentiment score of *-1* and a positive word with a score of *1*. The total sentiment score was determined by the sum of all of the negative and positive words found in the text of the tweet (Bing, Minqing, and Junsheng 2005).

```
twinklempatell: Just saw a mother at Walmart slap
the shit out of her daughter Bc she wouldn't stop crying.
Absolutely ridiculous.
```

Listing 3.3. Negative tweet example

The example in listing 3.3, the total sentiment score is *-3*. *slap*, *shit* and *crying* all yields a sentiment score of *-1* and their total sum is *-3*.

Iterating over all tweets in the data set the sentiment score of each tweet was calculated matching its content with sentiment dictionaries.

To view sample data from the sentiment dictionaries, review the appendix B section of this thesis.

3.4.3 Data aggregation

The Twitter and financial data sets for each company were combined and aggregated on a per day basis, and thereafter stored in separate data sets. Entries on weekends were added to the next weekday as the stock market is closed during the weekend. Table 3.3 presents and describes each of the aggregated variables.

The following definitions were used when aggregating and preprocessing the data sets.

Definition 1: a tweet is distinguished as having a *heavy influence* when the user posting the tweet has over 200 000 followers.

Definition 2: a tweet is classified as positive with a score of 1 and as very positive if it has a sentiment score larger than 1.

Definition 3: a tweet is classified as negative with a score of -1 and as very negative if it has a sentiment score small than -1.

The need for classification of *heavy influencers* arose when questioning if the common user is as influential as the more popular user. The threshold of 200 000 followers were set after testing the heavy influencers attributes impact on the linear regression model.

Lowering the threshold decreased the attributes impact and increasing the threshold limited the number of users classified as *heavy influencers* too much.

The threshold of ± 2 to classify a tweet as *very positive/negative* was set by analyzing tweets manually and finding a representative value for this threshold.

Table 3.4 presents sample aggregated data from tweets containing the keyword *walmart*.

Table 3.3. Aggregated values and sample data from Walmart

Aggregated Value	Sample Data	Description
created_at	2015-03-24	The date of which the data was collected and posted on Twitter.
all_tweet_count	3144	The number of tweets posted containing the search keywords. In this case, *walmart* or *WMT* is contained in the tweet.
positive_score_percentage	60	The percentage of positive tweets.
very_positive_percentage	19	The percentage of very positive tweets.
very_negative_percentage	7	The percentage of very negative tweets.
heavy_influence_count	157	The number of heavy influential Twitter users.
heavy_positive_influence_score	41	The percentage of heavy influential positive tweets.
very_heavy_positive_influence_percentage	11	The percentage of heavy influential very positive tweets.
very_heavy_negative_influence_percentage	3	The percentage of heavy influential very negative tweets.

Table 3.4. Sample of aggregated Walmart data.

Aggregated value	Sample day 1	Sample day 2	Sample day 3
created_at	2015-03-25	2015-03-26	2015-03-27
all_tweet_count	18049	15029	11307
positive_score_percentage	72	66	62
very_positive_percentage	24	24	17
very_negative_percentage	5	7	8
heavy_influence_count	607	539	416
heavy_positive_influence_score	57	67	67
very_heavy_positive_influence_percentage	24	34	27
very_heavy_negative_influence_percentage	3	4	11

3.4.4 Input data

The complete aggregated data contains the sentiment analysis score for each company and day combined as described in 3.4. Furthermore the financial data entry *open* as described in table 3.1 was added to the input data set for each day.

The input data for the classifiers were all the parameters presented in this table but the created_at parameter. This parameter was removed when training the classifiers as the interest was to learn from historical patterns and to predict the stock close price movement on a daily basis, as this approach have shown promising results in previous research (Makrehchi, Shah, and Liao 2013).

Furthermore, the parameter *direction* was added to the input data set which represents the stock close price movement for the recorded day as *up* or *down*. This parameter was used as the label parameter.

All of the input parameters were classified as *integer* values but the *direction* parameter.

3.4.5 Regression Analysis

Previous research have found that using a multiple regression analysis on stock variables such as open, close, and high price of the month, a model with a 89% accuracy on predicting stock price movement was established (Kamley, Jaloree, and Thakur 2013). Furthermore, researchers have found a significant correlation when using regression techniques between news values and weekly stock price changes at the beginning of each week (Yue Xu 2012).

The implemented multiple linear regression analysis is an least square regression model. The response variable is the close price variable being predicted by the remaining input variables serving as explanatory variables.

3.4.6 Classifier training

A split-validation approach was used to train the classifiers using subsets of the original data set for training and testing. The training and validation set size ratio was 80/20% as proposed sufficient in earlier research (Guyon 1997). The subsets were built using stratified sampling to ensure equal class distribution as in the original data set.

A 10-fold cross validation approach was used on the training data set in order to estimate the accuracy of the training model. Using this approach the data set is split into subsets where each subset is used exactly once for validation. Cross-validation is sub-optimal due to the low sampling variance but generally performs well (Esbensena and Geladib 2010).

The classifiers are evaluated analyzing the commonly used *accuracy*, *precision* and *recall* performance metrics (Hossin et al. 2011). Using these performance metrics we were able to optimize the classifiers at a training stage. The primary performance metric used for evaluation was the accuracy metric.

These basic performance metrics suffer from a number of limitations that could lead to suboptimal solutions (Hossin et al. 2011). However, they are easy to calculate and serve as traditional and reliable performance metrics. Furthermore they are commonly used in similar applications (Makrehchi, Shah, and Liao 2013; Paltoglou and Thelwall 2012).

The trained classifiers are also compared in a receiver operating characteristic (ROC) curve for visual evaluation. The *y-axis* of the curve represents the *true positive rate* whereas the *x-axis* is the corresponding *false positive rate*.

All of the supervised classifiers were configured to predict the outcome of the *direction* variable.

3.4.6.1 Accuracy

Accuracy is used to statistically measure the correctly identified classifications by a model. The following equation describes how *accuracy* was calculated.

$$Accuracy = \frac{\Sigma\text{true positives}+\Sigma\text{true negatives}}{\Sigma\text{true positives}+\Sigma\text{true negatives}+\Sigma\text{false positives}+\Sigma\text{false negatives}}$$

3.4.6.2 Precision

Precision is the number of correct classification predictions divided by the number of total predictions. *Precision* describes the percentage of positive predictions that were correct.The following equation describes how *precision* was calculated.

$$Precision = \frac{\Sigma \text{true positives}}{\Sigma \text{true positives} + \Sigma \text{false positive}}$$

3.4.6.3 Recall

Recall is the number of correct classification predictions divided by the total true number of correct classifications. *Recall* describes the percentage of positive cases that were identified by the classifier. The following equation describes how *recall* was calculated.

$$Recall = \frac{\Sigma \text{true positives}}{\Sigma \text{true positives} + \Sigma \text{false negatives}}$$

3.4.7 Naive Bayes

Similar approaches of using Naive Bayes as a classifier with sentiment analysis have been proven to be reliable, robust and accurate when analysing reviews (Paltoglou and Thelwall 2012).

The Naive Bayes implementation naturally applies the Bayes' theorem on the input variables. To prevent high influence of zero probabilities, Laplace correction was used. Laplace correction is the process of avoiding zero probabilities by adding one to each variable. This processes has a small impact on the estimated probabilities, as the data set size is large enough not to be influenced.

3.4.8 Support Vector Machine

Previous research on using SVMs for stock market forecasting have shown good accuracy, which increases as time span becomes longer. When compared to a basic linear regression, a generalized linear model and a baseline predictor model, the SVM model outperformed the other models (Shen, Jiang, and T. Zhang 2012). SVMs have been further proven to outperform other models such as Explanation-Based Neural Networks, Random Walk, Linear discriminant analysis and Quadratic Discriminant Analysis (W. Huang, Nakamori, and Wang 2005)

The implemented SVM classifier used the gamma kernel type of radial for stock close price movement using the other input variables.

3.4.9 Decision Tree & Random Tree

When predicting daily trends, the accuracy of decision trees, more specifically Multiple Additive Regression Trees (MARTs) have shown to reach a high accuracy of 74%, and are not as dependent and sensitive to the size of the training data as SVMs are (Shen, Jiang, and T. Zhang 2012). Because of this promising result, both

decision tree and random tree classification models have been trained and applied to the data.

The implemented decision tree and random tree classifiers used the criterion of information gain as favored in earlier research (Harris 2001) for splitting. The minimal gain for splitting was set to 0.1.

The tree was generated with pruning and prepruning. The model generated three prepruning alternatives if splitting on the selected node did not add enough discriminative power. The pruning confidence was set to 0.25.

The tuning variables were set after optimization on the accuracy metric using linear scaling with fixed steps on each variable.

3.4.10 Artificial Neural Network

The use of ANNs in financial forecasting is extensive (Kaastra and Boyd 1996) and have shown promising results in earlier research (Schoeneburg 1990; Olatunji et al. 2011; Das and Shorif Uddin 2011; Ticknor 2013). Therefore, an ANN implementation is of high interest for our application. Furthermore, ANNs have been proven to outperform *Statistical techniques* in stock market forecasting (Kunwar and Ashutosh 2010).

In order to predict the stock price movement we implemented a multi-layer perceptron feed-forward artificial neural network trained by a back propagation algorithm.

Artificial Neural Networks are subjects to optimization and tuning of the parameter settings to achieve optimal performance (Kaastra and Boyd 1996). Optimization on the parameters *training cycles, learning rate, momentum* and *decay* was performed using on a linear scale with a fixed step range.

The optimization criterion was to maximize the accuracy on the training set. This was achieved by evaluating the ANN accuracy on all of the possible tuning parameter combinations of the parameter settings presented in table 3.5.

Table 3.5. Attribute optimization settings.

Attribute	Min	Max	Steps
Training Cycles	100	3000	5
Learning Rate	0.1	1.0	10
Momentum	0.0	1	10
Decay	True/False		

The hidden layer and sigmoid size of

$$\frac{\Sigma\text{number of attributes}+\Sigma\text{number of classes}}{2}+1$$

were used as recommended by RapidMiner (RapidMiner 2015). To evaluate the optimality of these numbers various numbers of hidden layers and sigmoid sizes were used. These tests shown no increase in performance but equal or worse.

In order to make use of the Artificial Neural Network all the data was normalized using range transformation to a scale of $[-1, 1]$.

Chapter 4

Results

This chapter will provide the results that has been found in the collected tweet and stock data, with the ML techniques applied to them. The results are presented both in tabular and graphical form together with explanations of the results.

4.1 Regression Analysis

Aggregating the data for all of the three companies and computing a general regression analysis on the close price variable using the remaining input variables as explanatory variables, the results presented in table 4.1 were achieved.

As seen in table 4.1, the R^2 coefficient, also known as the sum of squares, which describes the goodness of fit of the model is close to 1 and the model therefore fits the data well. The R^2 value of 0.9993 implies that 99.9% of the cause for the stock close price are due to the explanatory input variables described in the method chapter. This is mainly due to the high significance and correlation of the opening price coefficient.

As seen in 4.1 in column $Pr(> |t|)$ representing the variable p-values describing the probability of the variable not being relevant, all of the twitter data variables show low level of significance and hardly contribute to the model, implying low levels of correlation.

The p-value significance threshold α is most commonly set to 0.05 (5%), implying no statistical significance for any of the twitter data attributes.

Very positive tweets from heavy influencers is the most significant twitter variable when predicting the stock close prise. The high p-value of 0.270 must still be considered, being significantly larger than the set significance threshold and furthermore implying a 27% probability of the variable not being relevant.

The standard error of the coefficient estimate measures the variability of the estimates. This error vary greatly in size of the ratio between the standard error and the coefficient estimate of the input variables. The only coefficient with a low standard error in comparison to the estimate is the open variable, implying heavy estimate variability in the twitter data variables.

Interesting findings in the linear model is the negative impact of positive tweets and score to the estimation of the stock close price as described in the *estimate* column in table 4.1. As previously mentioned, the significance of these variables is very low and the variables should therefore not be used as a predictor of the stock close price, but rather serve as unexpected findings.

Figure 4.1 shows the residuals of the linear model where it is clear that there are some heavy outliers from the models prediction implying high variance. This would be typical for all stock prediction models as the stock market takes heavy unexpected turns by nature.

Table 4.1. Summary of full data set linear regression model.

Residuals:				
Min	1Q	Median	3Q	Max
-17.9886	-1.1770	-0.1969	1.3623	11.0742

Coefficients:	Estimate	Std. Error	Pr(>|t|)
(Intercept)	4.615474	8.727520	0.601
Score	-0.013141	0.176264	0.941
Open	1.003223	0.006042	<2e-16
Very Pos Percentage	-0.141712	0.325940	0.667
Very Neg Percentage	-0.009787	0.550579	0.986
Heavy Score	-0.066294	0.080536	0.417
Heavy Very Pos Percentage	0.184323	0.163724	0.270
Heavy Very Neg Percentage	-0.051713	0.115817	0.659

Multiple R-squared:	0.9993
p-value:	<2.2e-16

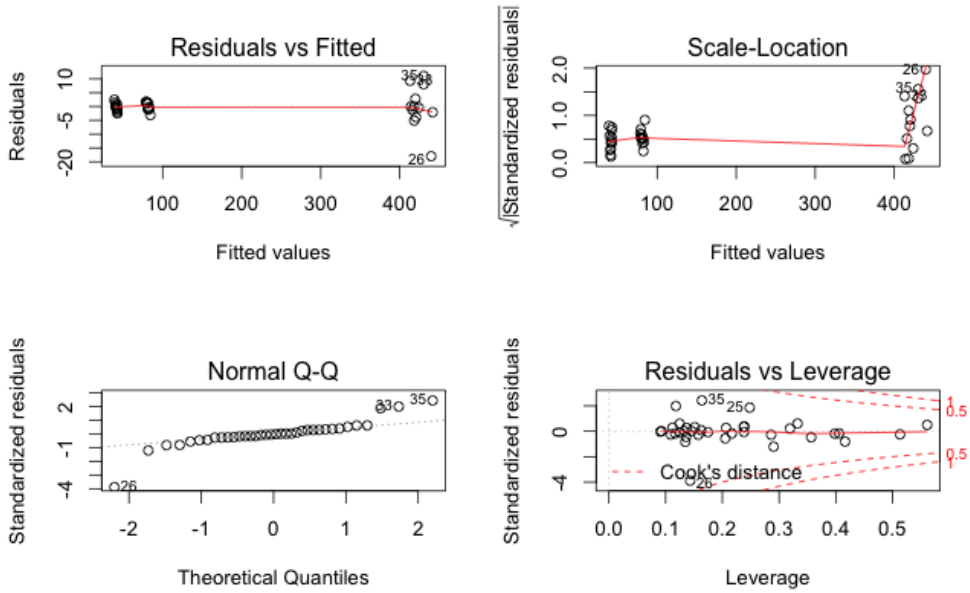

Figure 4.1. Linear model of full data set residuals.

The validity of the general regression analysis on company-specific prediction varied in result. Using the regression analysis results a prediction of the stock close price was conducted. These results are presented in figure 4.2, presenting the predicted close price, the actual close price and the opening price.

As seen in figure 4.3 the aggregated mean error of Microsoft is much larger than the mean error of Netflix on the predicted close price in comparison to the actual close price value using the general regression analysis.

Conducting a regression analysis on each company's specific data and applying the results to predict that company's stock close price is of high interest as the general regression analysis model might deviate.

The result of the company-specific regression analyses are presented in table 4.2, 4.3 and 4.4.

These results are interesting as they suggest much variety in variable relevance. The variable relevance for the general model presented in table 4.1 suggested that the only *quite* relevant coefficient is the *Heavy Very Pos Percentage* variable. This variable is highly relevant in the Walmart specific model as well as the Netflix specific model. Furthermore this variable is less relevant in the Microsoft specific model than the general model.

The relevance of the *heavy influencer* variables in the Walmart specific model in table 4.2 suggests that all of these are highly relevant as the variable p-values are smaller or slightly higher than the earlier mentioned p-value significance threshold.

The *Heavy Very Pos Percentage* coefficient is even more relevant for the Netflix specific model.

The R^2 for the company-specific models suggests that model fit is best for Walmart, followed by Netflix and last Microsoft. This assumption is further presented in figure 4.4 describing the mean prediction error for each company when using specific company data in the regression analysis. Is it obvious that the company-specific model outperforms the general model and offers promising results.

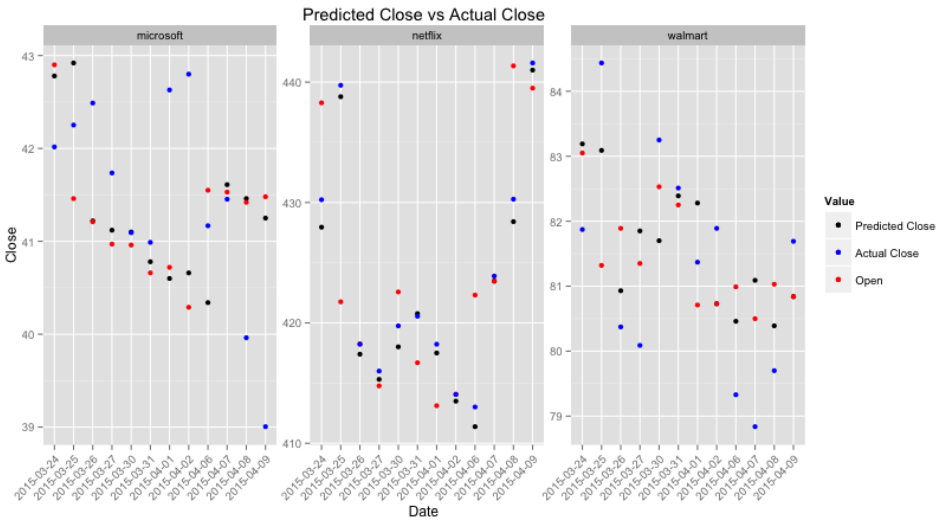

Figure 4.2. Predicted Close vs True Close using the full data set for the regression analysis.

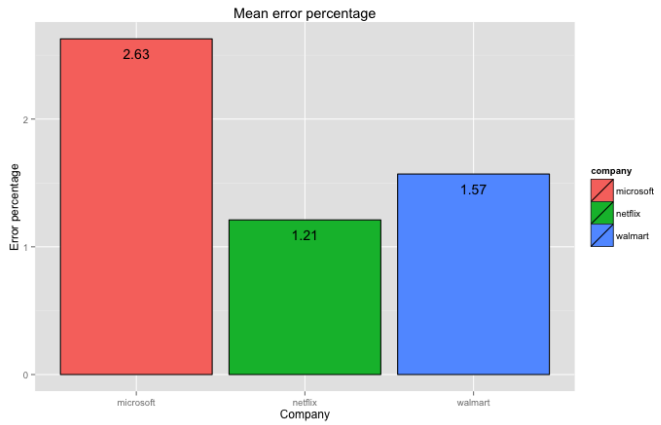

Figure 4.3. Prediction mean error percentage per company using the full data set for the regression analysis.

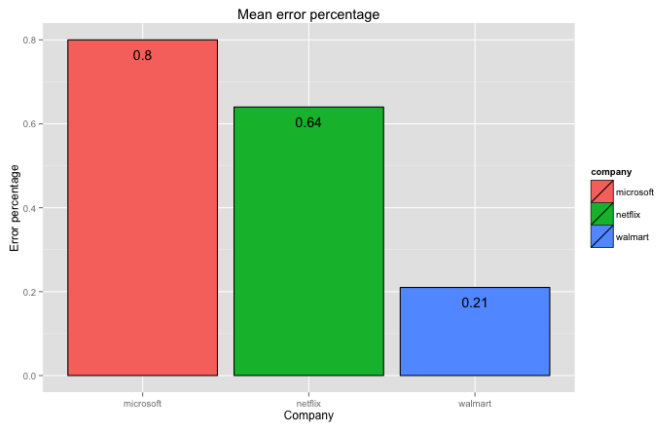

Figure 4.4. Prediction mean error percentage using company-specific regression analysis results.

4.2 Supervised learning

Classification on the label *direction* described in the method chapter using the entire data set produced the results of the stock close price movements prediction presented in table 4.5. These results show a great variation in quality of the classifiers in terms of accuracy, precision and recall rate.

Figure 4.5 presents the ROC chart for the used classifiers and provides a graphical overview of the performance for these. Further investigation on the Random

Table 4.2. Summary of Walmart specific data set linear regression model

Coefficients:	Estimate	Std. Error	Pr(>|t|)
(Intercept)	62.197785	11.238309	0.00264
Score	0.024919	0.022324	0.31507
Open	0.232718	0.138790	0.15444
Very Pos Percentage	0.021648	0.033405	0.54552
Very Neg Percentage	0.006328	0.106493	0.95492
Heavy Score	-0.019271	0.007792	0.05631
Heavy Very Pos Percentage	-0.051990	0.019922	0.04769
Heavy Very Neg Percentage	-0.027675	0.008767	0.02519

Multiple R-squared:	0.9205
p-value:	0.01672

Table 4.3. Summary of Netflix specific data set linear regression model

Coefficients:	Estimate	Std. Error	Pr(>|t|)
(Intercept)	120.8096	86.3701	0.2208
Score	0.4920	0.7901	0.5607
Open	0.7096	0.1920	0.0141
Very Pos Percentage	-1.0216	1.0796	0.3875
Very Neg Percentage	0.5552	2.6090	0.8399
Heavy Score	-0.4811	0.3163	0.1888
Heavy Very Pos Percentage	1.6575	0.5442	0.0286
Heavy Very Neg Percentage	-1.8312	1.2763	0.2108

Multiple R-squared:	0.8977
p-value:	0.01672

Tree and Decision Tree show that they both suffer greatly from overfitting and do therefore not provide general performance, as could be assumed from viewing the chart. This could also be the case for the ANN, however the probably is low as the number of hidden layers and nodes are low (Panchal et al. 2011).

From the results in table 4.5 it can be seen that the ANN serves as the highest performing classifier on the general data set containing information from all three companies.

Table 4.4. Summary of Microsoft specific data set linear regression model

Coefficients:	Estimate	Std. Error	Pr(>|t|)
(Intercept)	27.389280	12.810523	0.0855
Score	0.003147	0.042580	0.9440
Open	0.429124	0.346484	0.2705
Very Pos Percentage	-0.065458	0.070711	0.3971
Very Neg Percentage	-0.070858	0.140564	0.6356
Heavy Score	-0.027609	0.072596	0.7193
Heavy Very Pos Percentage	-0.033201	0.044483	0.4890
Heavy Very Neg Percentage	-0.142004	0.112673	0.2632

Multiple R-squared:	0.6971
p-value:	0.03012

Table 4.5. Supervised learning algorithm results on full data set.

Method	Accuracy	Precision (Up)	Precision (Down)	Recall (Up)	Recall (Down)
Naive Bayes	33%	41%	26%	33%	33%
SVM	52%	56%	24%	86%	7%
Decision Tree	55%	61%	46%	67%	40%
Random Tree	53%	58%	40%	71%	27%
Artificial Neural Network	68%	70%	62%	76%	53%

4.2.1 ANN

The performance of the ANN varied using different settings, but most settings outperformed other supervised classifiers in terms of accuracy, precision and recall. *Parameter Optimization* as described in the method chapter on *training cycles, learning rate, momentum* and *decay* resulted in optimized parameter settings presented in table 4.6.

Table 4.6. Optimized settings of the multi-layered back propagation training algorithm.

Training Cycles	Learning Rate	Momentum	Decay
2420	0.82	0.5	False

Given parameter settings increased the performance of the neural network in comparison to the initial parameter settings for the general data set. The optimized parameter performance and the area under the curve (AUC) is presented in table 4.7. Various numbers of hidden layers and nodes were also tested but the best performance was achieved using the algorithm as described in subsection 3.4.10 resulting in one hidden layer with four nodes.

Given the assumption that the stock price movement is either *up* or *down* the accuracy of the random guess is 50%. The optimized ANN performs well on the data set and provides a more robust prediction of stock price movement. As previously

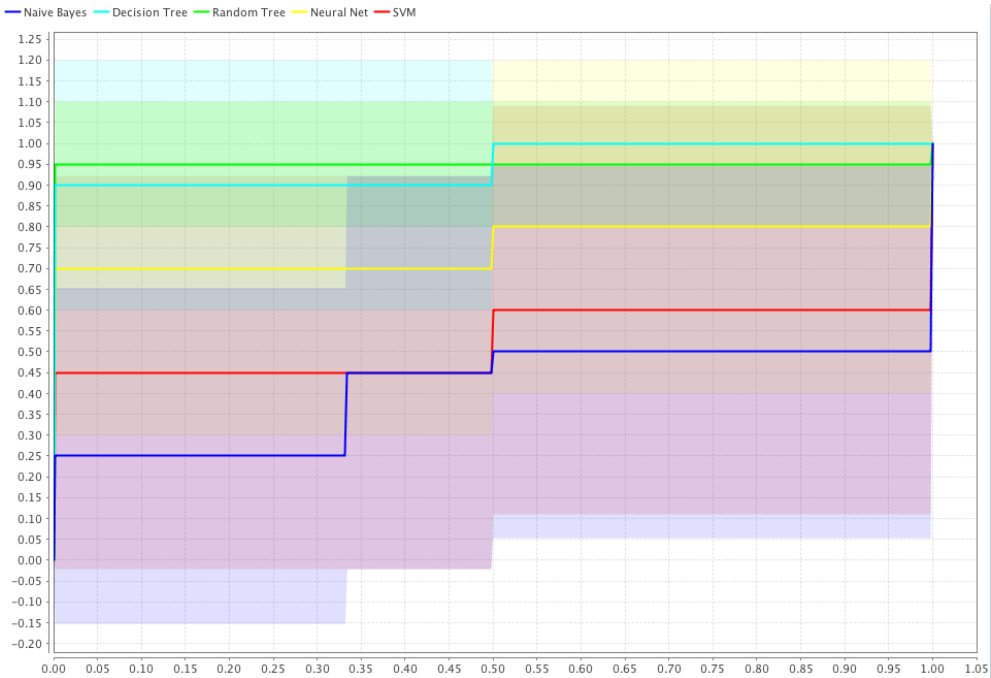

Figure 4.5. Receiver operating characteristic chart of given results.

Table 4.7. ANN performance on the full data set with optimized parameters.

Method	Accuracy	Precision (Up)	Precision (Down)	Recall (Up)	Recall (Down)	AUC
Artificial Neural Network	76%	71%	80%	83%	33%	0.867

mentioned this might be a result of *overfitting* and therefore not applicable with the same accuracy to other sets of data.

If the given accuracy is *good enough* for a real-world application is arguable. Furthermore investors try to maximize the potential profit and would therefore be more interested in actual stock close price value rather than non-specified stock close price movements.

However, with the given 76% accuracy of the ANN it is possible to make predictions of the movement with a 52% higher accuracy than the 50% accuracy of the random guess.

As earlier mentioned the ANN only predicts the movement rather than the percentage value of the movement. Considering this constraint in combination with brokerage fees it is not possible to place investments with a good profit margin at a high rate of certainty based on the ANN classification.

As an example, purchasing stocks given price movement prediction of *up* the actual stock price increase might be 0.1% not covering the brokerage fee of 0.25% (Skandiabanken 2015) resulting in a margin loss of 0.15%.

As companies might be influenced more or less by social media it is of high interest to train the ANN using company-specific data in order to increase performance.

The results from the company-specific classification predictions on stock price movement are presented in table 4.8. These results suggest that company-specific classification only outperformed the general classifier for the company Walmart in terms of the evaluated performance metrics as described in the method chapter.

Furthermore, these results are in-line with the given results by the company-specific linear regression model where Walmart had the lowest mean prediction error, followed by Netflix and last Microsoft as presented in figure 4.4.

Table 4.8. ANN performance on the company-specific data sets with optimized parameter settings.

Company	Accuracy	Precision (Up)	Precision (Down)	Recall (Up)	Recall (Down)
Walmart	80%	100%	71.43%	71.43%	100%
Netflix	60%	57.14%	60%	66.67%	100%
Microsoft	55%	63.64%	0%	87.5%	0%

Chapter 5

Discussion

This chapter will present an analysis of the results, discussion about the limitations, methodical constraints together with a conclusion and future work of this thesis. The implementational and computational limitations are discussed with focus on restrictions on time, data quantity and machine learning implementations. Finally, the conclusion of the found results are discussed with advice of future research in the areas.

5.1 Analysis of results

The found results propose that most classification models do not yield satisfiable performance predicting stock price movements. In order to find more accurate predictions, consulting ANN methods is necessary. The implemented ANN was the best performing classifier in terms of our evaluated performance metrics and offered good accuracy for stock price movement.

The accuracy of the trained classifiers are all constrained by the low amount of available data and are subjects to overfitting.

As seen in table 4.5, the worst performing classifier was *Naive Bayes*. These results are surprising, as previously mentioned in the background, that *Naive Bayes* have previously been proved to be optimal no matter how strong the dependencies among the attributes are, if the dependencies distribute evenly in classes or if they cancel each other out (H. Zhang 2004). This is arguably due to the restricted amount of data, discussed in the next section.

Evaluating figure 4.3, the mean error of Microsoft was the largest, whereas Netflix had the lowest. This could arguably be due to the size of Microsoft's organization. Microsoft is an international, multi-million corporation and their stock is affected by various of volatilises in the world of stock trading. Still, this is surprising as larger companies tend to have a more stable stock price and would therefore be more suitable for statistical prediction.

Analyzing the company-specific regression analysis mean error it is obvious that the predictability of the stock close price using social media data variables varies.

This is visualized in the prediction mean error graph for the company-specific regression model presented in figure 4.4. Furthermore this theory is enhanced by the performance of the company-specific ANN presented in table 4.8. The linear regression and ANN predictability of a company's stock both perform best when predicting the Walmart stock, followed by the Netflix stock and last the Microsoft stock.

Furthermore, the data gathering could be the biggest reason of error in this analysis, since only keywords such as *Microsoft* and *MSFT* were gathered from the Twitter streaming feed, not accounting for any sub-organizations and products. In conclusion, the gathered data could have been too narrow in order to create an effective analysis of the entire corporation.

Our findings suggests that the public opinion concerning a company *do not* alter the stock effectively neither in a positive nor negative way. Only twitter accounts with more than 200 000 followers have an impact, positively and negatively. Such accounts are often news sources and reporters reporting on company specific news, leaks and events.

Our results suggest that the use of Twitter sentiment analysis as an exclusive stock market predictor is not reliable enough to be a used in a real-world application. However, it provides an extra layer of predictability as a support tool to an existing stock market prediction system.

5.2 Limitations

The stock market is volatile by nature and is much affected by global factors, such as economical, political, social and technological. The methodological constraints on this thesis consists primarily of *time*, *computational power* and knowledge in the field of *Machine Learning* and *Sentiment Analysis*.

The results of this thesis add to previous empirical results the importance of big data, a complete sentiment analysis and the significance of using artificial neural networks when predicting stock prices with the help of social media posts. By using statistical machine learning, collecting large amounts of data in a longer period of time is necessary in order to create predictions with higher accuracy.

Our work was restricted by limited tweet data and a non-complete sentiment analysis. The use of emoticons, emojis and slang on Twitter is popular and the used sentiment analysis dictionary did not account for these aspects in a complete manner. Furthermore, the context of the tweets were not taken in consideration.

The limitations of time, data, computational power and knowledge of ML has formed a major drawback, since this research has been limited to analyze specific stocks over a short period of time with low data quantity and limited AI and ML knowledge.

Twitter do not provide the availability to search and gather historical data older than seven days and the only way to retrieve older data sets are to purchase them. This thesis has therefore been limited to gather future data which is the primary

reason for the low quantity of data. To ensure data integrity and eliminate the potential risk of altered data, we made the choice to collect the data ourselves using the Twitter Streaming API.

5.3 Conclusion

In order to achieve more valid results there is a need for larger amounts of data. The currently limited amount of twitter data restricts the valditiy of the used machine learning methods and do not provide results reliable enough to be exclusively used in a real-world application.

The implemented sentiment analysis dictionaries were not analyzed carefully and the threshold of the inter-rater reliability of 79-80%, mentioned in section 2.3.1, was not taken into account when choosing this method. With a more foolproof and complete implementation of SA taking emoticons, emoijis, slang and context into consideration the accuracy of the predictions from the ML models might be enhanced.

The optimized implementation of the feed-forward neural network outperformed other types of machine learning techniques with relatively high performance and accuracy. However, this accuracy is limited to stock price movement rather than stock close price prediction.

We can conclude that solely, the common users voice on twitter do not impact the stock price movement much, if any at all, but the heavy influencers' positive and negative feedback did have an impact.

In terms of correlation and causality, as discussed in chapter two, the found results cannot be classified into any of the four cases that exist in correlation theory with certainty. Our conclusion is that *there exists a weak relationship* between a companies stock and their respective social media posts. But if this relationship is strong enough to be classified as a correlation or is a subject to low data quantity and overfitting, is debatable.

The use of Twitter sentiment analysis as a stock predictor is not reliable enough to be used as a exclusive predictor. This approach to stock market prediction serves better as an extra layer of complexity, potentially adding accuracy to an existing implementation, considering the relatively high accuracy on stock price movement achieved by the ANN.

5.4 Future research

Future research in the field could investigate the importance of further developing the sentiment analysis to take more parameters in consideration as described in the conclusion. When analyzing social media, new trends such as the use of emoticons, emojis and language slang must be taken into account in order to get satisfiable accuracy of the sentiment analysis (Gonçalves, Benevenuto, and Cha 2013).

Furthermore, gathering of social media data during a longer period of time would be of interest. Extending the data mining to gather information from financial resources and newspapers could serve as an extension to traditional stock market prediction approaches on financial data only.

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Appendix A

Twitter Keywords

The following search terms were used to collect the data from Twitter. The search words for each company is the company name and their respective name at the stock market.

Microsoft

- Microsoft
- MSFT

Netflix

- Netflix
- NFLX

Walmart

- Walmart
- WMT

Appendix B

Sentiment Analysis Dictionary Samples

Table B.1 contain examples of **positive** and **negative** words that might be found in the English language and tweets (Bing, Minqing, and Junsheng 2005).

positive	**negative**
accomplish	angry
admire	attack
blessing	betray
ecstasy	bias
energize	bitch
fantastic	cancer
good	dead
kudos	flaw
like	hate
smile	insane

Table B.1. Examples of positive and negative words.